03-26 第2篇

阅读理解（说明文） How Fireworks Get Their Colours

① On Spring Festival night, millions of Chinese families look up at the same beautiful sight — the sky full of **red, green, gold, and purple flowers**. A firework lasts only a few seconds, yet behind it lies hundreds of years of science. Have you ever wondered **why different fireworks have different colours, and how these colours are painted so quickly in the black sky?**

② To answer this, we must first understand the **"skeleton" of a firework**. A typical festival firework has three main parts: a **paper tube** on the bottom, a small **lifting charge** of black powder that pushes the tube into the sky, and a round **"star ball"** inside. The star ball is a tight group of small balls called **stars**. Each star is about the size of a red bean and is made of three things: a **fuel** to burn hot, an **oxidiser** to feed the flame, and a small amount of **metal salt** to make the colour.

③ So where do the colours come from? The secret is the metal salts. When a certain metal is heated to a very high temperature, it gives out **light of its own special colour** — this is what scientists call an **"atomic signature"**. Different metals have different signatures:

Colour	Main metal salt
Red	Strontium (Sr) salts
Orange	Calcium (Ca) salts
Yellow	Sodium (Na) salts
Green	Barium (Ba) salts
Blue	Copper (Cu) salts
Purple	A mix of copper and strontium
Silver-white	Aluminium or titanium powder

So, one red peony in the sky is really a shower of **tiny burning red-bean stars**, each one filled with strontium.

④ Why does a firework open into such a neat flower pattern? It is because the stars are carefully placed inside the star ball, like **seeds in a round pumpkin**. When the star ball explodes in the sky, the explosion pushes the stars outwards at the same speed. They fly in all directions, burn at the same time, and our eyes see a quick round flower. Some skilled makers can even arrange the stars into hearts, letters, or smiley faces. In 2027, several cities in China are testing **drone fireworks** that add no metal salts but send

thousands of programmed lights into the sky — beautiful, quiet, and much cleaner for the air.

⑤ Yet, the old chemical firework is not only beautiful; it also reminds us to be careful. **Every year**, hospitals in China treat thousands of eye and hand injuries caused by unsafe fireworks. Experts advise teenagers to enjoy fireworks at a safe distance, never pick up a "dead" piece from the ground, and never play with small fireworks indoors. Technology is growing; it is our wisdom that must grow together with it.

根据文章内容，选择最佳答案。

1. What is the article mainly about?

A. How to make Spring Festival dinners more fun.
B. How fireworks produce different colours and patterns, with safety tips.
C. Why the Spring Festival is the most important holiday in China.
D. The history of black powder in ancient China.

2. What is inside the "star ball" of a firework?

A. A group of small "stars", each made of fuel, oxidiser, and a metal salt.
B. A special kind of silk ribbon that burns brightly.
C. A piece of bamboo that flies high in the sky.
D. A small lifting charge and nothing else.

3. According to the table in paragraph ③, which metal salt makes green fireworks?

A. Strontium salts.
B. Sodium salts.
C. Barium salts.
D. Copper salts.

4. Why does a firework explode into a round flower pattern?

A. The sky pulls the stars into a circle by magic.
B. The stars are placed carefully in the ball and fly outwards at the same speed.
C. The burning smoke always moves in circles.
D. A strong wind blows the stars into a round shape.

5. What is the writer's main message in the last paragraph?

A. Drone fireworks are the only acceptable kind today.
B. Everyone in China should stop watching fireworks.
C. Fireworks are beautiful, but safety and wisdom must go with technology.
D. Only adults can enjoy watching fireworks safely.

答案与解析

1.
【答案】B
【题型】主旨大意题
【定位】全文围绕烟花色彩原理、结构与安全提醒
【解析】主旨为"烟花色彩产生原理与安全建议"。A、C、D 非主题。故选 B。

2.
【答案】A
【题型】细节理解题
【定位】第②段 "small balls called stars... fuel, oxidiser, metal salt"
【解析】"星球"由许多星星组成，每颗含燃料/助燃剂/金属盐。B、C、D 错。故选 A。

3.
【答案】C
【题型】表格定位题
【定位】第③段表格 Green — Barium salts
【解析】绿色由钡盐产生。A、B、D 对应其他颜色。故选 C。

4.
【答案】B
【题型】细节理解题
【定位】第④段 "stars are carefully placed... pushes outwards at the same speed"
【解析】事前排列，等速飞散形成圆花。A、C、D 无依据。故选 B。

5.
【答案】C
【题型】观点态度题
【定位】第⑤段 "Technology is growing; it is our wisdom that must grow together with it"
【解析】欣赏美丽同时强调安全与智慧。A、B、D 过度或错误。故选 C。

词汇注释（超纲词）

- **strontium** [n.] 锶（金属元素）
- **barium** [n.] 钡（金属元素）
- **oxidiser** [n.] 氧化剂